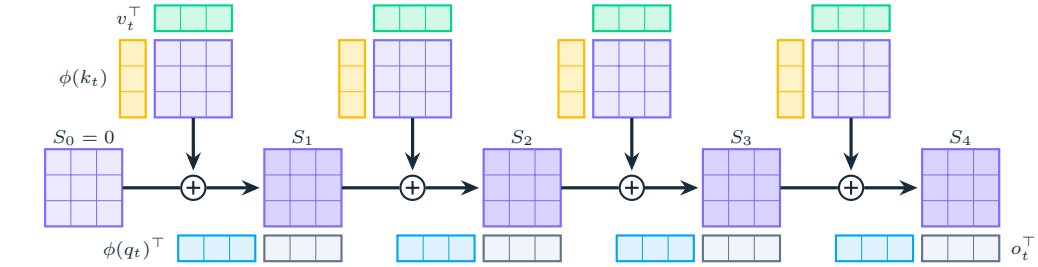


Softmax attention keeps every key and value

$$o_t = \sum_{j=1}^{t} \frac{e^{q_t^\top k_j / \sqrt{d}}}{\sum_{l=1}^{t} e^{q_t^\top k_l / \sqrt{d}}} v_j$$

step t reads $2t$ vectors,
about $4td$ operations
cache: $2td$ numbers, growing with t



Linear attention keeps one $r \times d$ matrix

$$S_t = S_{t-1} + \phi(k_t)\, v_t^\top$$
$$z_t = z_{t-1} + \phi(k_t)$$
$$o_t = S_t^\top \phi(q_t) \,/\, z_t^\top \phi(q_t)$$

each write is a rank-one $r \times d$ matrix
step t: about $4rd$ operations
state: $rd + r$ numbers, the same for every t
the drawing leaves out z_t and the division

One head with $d = r = 128$, after $t = 131{,}072$ tokens: the cache holds $33{,}554{,}432$ numbers and the state $16{,}512$, $2{,}032$ times fewer. The state is the size of the cache after about 64 tokens.